

Principal Component Analysis of Indian Equity Returns

FACTOR DISCOVERY & COVID-19 STRUCTURAL BREAK

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20 April 2026

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Bachelor of Statistics (Hons.) 2024–27

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Part I: What is a factor?

What is a factor?

Definition

A **factor** is a hidden, systematic force that drives the returns of many assets simultaneously. It is *latent* — we do not observe it directly; we infer its influence from co-movement.

Everyday analogy

Weather is a factor for umbrella sales, farm output, and tourism — one hidden force, many observable outcomes.

Finance analogy

An RBI rate surprise simultaneously affects:

- ▷ all PSU bank stocks
- ▷ housing-finance NBFCs
- ▷ bond-sensitive utilities

The linear factor model

$$r_i = \alpha_i + \sum_{k=1}^K b_{ik} f_k + \varepsilon_i$$

- f_k — common factors (unobserved)
- b_{ik} — stock i 's *loading* on factor k
- ε_i — idiosyncratic (stock-specific) noise

Key insight: If factors are few but stocks are many, then the enormous covariance matrix Σ is *compressible* — most dependence lives in a low-dimensional subspace.

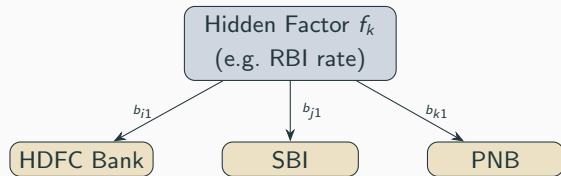
Factors in Investing — Economic Meaning

Why do factors drive returns?

Assets sharing exposure to the same *risk source* must offer the same *compensation* for that risk, otherwise arbitrage opportunities exist.

Concrete examples:

- **Revenue growth** factor: companies with accelerating revenue beat the cross-section in expansions.
- **Rupee depreciation** factor: IT exporters rally; oil importers fall — the USD/INR rate is the hidden force.
- **Crude oil** factor: energy and chemicals co-move; airlines diverge.



One factor f_1 (RBI surprise) simultaneously affects all PSU banks through their respective loadings.

From CAPM to Fama–French — A Historical Arc

CAPM (Sharpe, 1964): 1 factor

$$\mathbb{E}[r_i] = r_f + \beta_i(\mathbb{E}[r_m] - r_f)$$

A stock's expected return is fully explained by its market beta.

Problem: Cross-sectional variation in returns cannot be explained by β alone.

Fama–French 3-Factor (1993)

$$r_i - r_f = \alpha + \beta_m r_m + \beta_s \text{SMB} + \beta_v \text{HML} + \varepsilon$$

- **SMB** = Small Minus Big (size premium)
- **HML** = High Minus Low book-to-market (value premium)

The "Factor Zoo" (Harvey et al. 2016)

Over 300 factors have been published in academic finance: momentum, quality, low-volatility, profitability, investment, liquidity, . . .

The proliferation problem:

- Many factors are correlated with each other
- Many are artefacts of data-mining
- The "true" number of priced factors is unknown

Our approach: Use PCA — a *data-driven* method to discover how many independent common directions actually exist in the Indian equity cross-section, without pre-specifying which factors to test.

Part II: Statistical Toolkit

PCA — The Core Idea

What PCA does

Given data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$ (mean-centred), PCA finds unit directions $\mathbf{w}_k \in \mathbb{R}^D$ that *maximise explained variance* in sequence:

$$\mathbf{w}_1 = \arg \max_{\|\mathbf{w}\|=1} \mathbf{w}^\top \hat{\Sigma} \mathbf{w}$$

where $\hat{\Sigma} = \frac{1}{N-1} \mathbf{X}^\top \mathbf{X}$ is the sample covariance.

This is the eigenvalue problem:

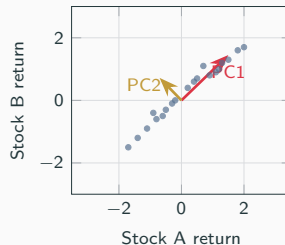
$$\hat{\Sigma} \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D \geq 0$$

In practice: compute via SVD $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$; numerically stabler than eigendecomposing $\hat{\Sigma}$ directly.

Fraction of variance explained (EVR):

$$\text{EVR}_K = \frac{\sum_{k=1}^K \lambda_k}{\sum_{k=1}^D \lambda_k}$$

Correlated returns $\rightarrow$ PCA finds axes



Dimensionality Reduction & the Factor Model

The fundamental decomposition

$$\underbrace{\Sigma}_{\substack{D \times D \\ \text{full covariance}}} = \underbrace{\mathbf{B} \Sigma_f \mathbf{B}^T}_{\substack{\text{low-rank common} \\ \text{component}}} + \underbrace{\Psi}_{\substack{\text{diagonal} \\ \text{idiosyncratic}}}$$

Parameter count comparison:

Model	Parameters	$D = 118$
Full $\hat{\Sigma}$	$D(D+1)/2$	7,021
Factor model ($K = 5$)	$DK + K^2 + D$	731

A **10× compression** — and more stable estimates because fewer parameters are estimated from the same data.

Why PCA is the natural estimator:

PCA on $\hat{\Sigma}$ recovers the loading matrix $\mathbf{B}$ under the factor model when the number of assets $D \rightarrow \infty$ (Connor & Korajczyk, 1988).

Crucially, we do **not** need to name the factors in advance — the data tell us how many independent directions of co-movement exist.

Marchenko–Pastur Law — Separating Signal from Noise

The problem

PCA *always* returns D components — even from pure noise. So which eigenvalues represent genuine common factors, and which are artefacts of finite-sample estimation?

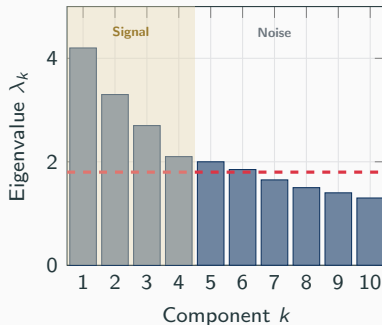
The MP bound

For an $N \times D$ matrix of i.i.d. noise with variance σ^2 , random matrix theory predicts the largest eigenvalue concentrates around:

$$\lambda_+^{\text{MP}} = \sigma^2 \left(1 + \sqrt{\frac{D}{N}} \right)^2$$

Any eigenvalue *exceeding* λ_+^{MP} is statistically inconsistent with pure noise.

Scree plot: signal vs. noise



For our daily panel ($N/D \approx 23.6$), the MP bound is relatively low, allowing more components to clear it.

Cross-Sectional Standardisation — Why and How

The problem without standardisation

Raw equity returns have very different volatilities across stocks. A high-vol small-cap will dominate PCA simply by having large variance — not because it carries more factor information.

More importantly: if all stocks move with the market today, that correlation should not dominate our factors. We want *relative* co-movement.

The transformation

At each time period t , standardise across the cross-section of D stocks:

$$\tilde{r}_{i,t} = \frac{r_{i,t} - \bar{r}_t}{\sigma_t^\times}, \quad \bar{r}_t = \frac{1}{D} \sum_i r_{i,t},$$

$$\sigma_t^\times = \sqrt{\frac{1}{D-1} \sum_i (r_{i,t} - \bar{r}_t)^2}$$

What it achieves

- Removes the contemporaneous *market component* from every row
- PCA then operates on **style and sector co-movements**, not on broad market beta
- PC1 explaining $\sim 4.7\%$ is a *feature*, not a deficiency — it reflects richer, multi-factor structure once the market-wide piece is stripped out

Rolling PCA uses a different standardisation: within-window time-series standardisation, so that changes in *market-wide co-movement level* remain visible for structural break analysis.

Component Selection — How Many PCs are Real?

Three complementary criteria:

[A] Scree Plot

Plot λ_k vs. k . Look for an **elbow** — the point where the rate of decline flattens. Components before the elbow carry signal; those after resemble noise.

Limitation: Subjective; elbow may be gradual.

[B] MP Threshold

Retain all k with $\lambda_k > \lambda_+^{\text{MP}}$. Provides a rigorous statistical threshold grounded in random matrix theory.

Limitation: Depends on the N/D ratio; can be too conservative for rolling windows with low N/D .

[C] Cumulative EVR

Choose the smallest K such that $\text{EVR}_K \geq 70\%$ (or another threshold). Simple and interpretable.

Limitation: Threshold is arbitrary; not adaptive to noise level.

In this project: All three criteria are used jointly. The scree plots (daily and monthly) show no sharp elbow — consistent with a *multi-factor* structure with many weakly contributing style factors rather than one dominant component.

Part III: Time-Variation Methods

Why Static PCA is Insufficient

Static PCA assumes the factor structure is constant across the entire sample. This is implausible when the sample spans structurally distinct regimes.

Events in our 2015–2026 window:

- **Nov 2016** — Demonetisation: sudden shock to cash-heavy sectors
- **2018** — NBFC crisis: credit-linked co-movement spikes
- **Mar 2020** — COVID-19 crash: extreme synchronisation across all sectors
- **2022–24** — Post-COVID regime: changed sectoral leadership

A single PCA on the full sample gives you the *average* structure — but it cannot tell you whether the structure *before* COVID was the same as *after*.



Two remedies:

1. **Rolling PCA** — track factor *strength* over time
2. **Procrustes alignment** — compare factor *composition* across windows

Rolling PCA — Idea and Mechanics

Algorithm

1. Fix window width $W = 24$ months
2. For each window ending at month t :
form $\mathbf{X}_t \in \mathbb{R}^{W \times D}$ from the most recent W returns
3. Apply Ledoit–Wolf shrinkage PCA to $\mathbf{X}_t$
4. Record eigenvalues $\{\lambda_1(t), \dots\}$ and loadings
 $\mathbf{V}_t \in \mathbb{R}^{D \times K}$
5. Advance t by one month and repeat

What we track: The fraction of variance explained by PC1 at each t :

$$\rho_1(t) = \frac{\lambda_1(t)}{\sum_k \lambda_k(t)} \in (0, 1)$$

This is a **co-movement intensity indicator**.

Why $W = 24$ months?

Too short: Covariance estimates are noisy; PCA is unstable.

Too long: Structural changes get blurred; temporal resolution is lost.

$W = 24$ months balances **statistical precision** (enough observations) with **temporal resolution** (sensitive to COVID-level shocks).

Economic interpretation: Rising $\rho_1(t) \Rightarrow$ stocks are moving more in lockstep $\Rightarrow$ diversification becomes less effective.

Orthogonal Procrustes Alignment — Why It Is Needed

PCA loadings are not uniquely identified. They can differ by (a) an arbitrary sign flip, or (b) a rotation within the eigenspace when eigenvalues are close. Naïve comparison of $\mathbf{V}_{\text{pre}}$ vs. $\mathbf{V}_{\text{post}}$ is therefore *misleading*.

The room analogy

Two surveyors measure the same room but draw their axes in different directions. The room hasn't changed; their coordinate systems differ.

Procrustes alignment rotates one set of axes onto the other so we can judge whether the *room itself* changed — not just the labelling of directions.

Three sources of artefactual difference:

1. **Sign flip:** $\mathbf{v}$ and $-\mathbf{v}$ represent the same direction
2. **Rotation:** When $\lambda_k \approx \lambda_{k+1}$, any rotation within the span $\{\mathbf{v}_k, \mathbf{v}_{k+1}\}$ is equally valid
3. **Basis labelling:** Factor 1 in window t may correspond to Factor 2 in window $t + 12$

Orthogonal Procrustes — Mathematics

The optimisation problem

Given reference $\mathbf{V}_{\text{ref}} \in \mathbb{R}^{D \times K}$ and current $\mathbf{V}_t \in \mathbb{R}^{D \times K}$, find the orthogonal rotation $\mathbf{Q}^*$ that brings $\mathbf{V}_t$ closest to $\mathbf{V}_{\text{ref}}$:

$$\mathbf{Q}^* = \arg \min_{\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}} \|\mathbf{V}_t \mathbf{Q} - \mathbf{V}_{\text{ref}}\|_F$$

The constraint $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$ ensures we only *rotate/reflect*, never distort.

Closed-form solution via SVD

Compute $\mathbf{V}_{\text{ref}}^\top \mathbf{V}_t = \mathbf{U} \mathbf{S} \mathbf{V}^\top$. Then:

$$\mathbf{Q}^* = \mathbf{U} \mathbf{V}^\top$$

Procrustes disparity

After aligning $\tilde{\mathbf{V}}_t = \mathbf{V}_t \mathbf{Q}^*$:

$$\delta(t) = \frac{\|\mathbf{V}_{\text{ref}} - \tilde{\mathbf{V}}_t\|_F}{\sqrt{K}}$$

- $\delta = 0$: identical loading compositions
- $\delta \rightarrow 1$: factor structures are orthogonal

In this project: $\delta = 0.782$ between pre- and post-COVID loading matrices — indicating the *identity of the stocks driving systematic co-movement* changed very substantially around March 2020.

Chow Test for Structural Breaks

What it tests

We model the rolling PC1 variance series $\{y_t\}$ as a linear time trend:

$$y_t = \alpha + \beta t + \varepsilon_t$$

The Chow test asks: *did the regression coefficients (α, β) change at a specified break point?*

Test statistic

$$F = \frac{(RSS_R - RSS_U) / k}{RSS_U / (T - 2k)} \sim F(k, T - 2k) \text{ under } H_0$$

where $k = 2$ (intercept + slope), RSS_R = pooled residuals, $RSS_U = RSS_{\text{pre}} + RSS_{\text{post}}$.

Intuition

If the market regime is stable, the pre- and post-break linear trends should have the same slope and intercept. A large F means the two sub-sample regressions are *significantly different from each other* — evidence of a structural break.

Important distinction:

The Chow test detects a change in the **trend** (α, β) — not merely a change in the *level* of y_t . A directional reversal of the slope (pre-COVID declining → post-COVID rising) is the strongest possible evidence of structural change.

Part IV: Empirical Study

Dataset — Universe, Source, and Variables

Universe construction

- 145 NSE/BSE-listed equities; seed from NIFTY 150 (large-, mid-, small-cap)
- Price data: Yahoo Finance via `yfinance` (adjusted close — split- and dividend-adjusted)
- Full sample: Jan 2015 – Apr 2026 (≈ 11 years, 2,781 trading days)

Variables computed

- **Daily excess returns:** $\tilde{r}_{i,t} = r_{i,t} - r_f$, where $r_f = 6.5\%$ p.a. (91-day T-bill proxy) and $r_{i,t} = (P_{i,t} - P_{i,t-1})/P_{i,t-1}$
- **Monthly returns:** last-price resampling

Data quality audit

Criterion	Threshold
Missingness	$< 40\%$
Extreme daily returns	< 5 observations $> \pm 50\%$
Stale-price streak	No 10+ identical consecutive values
Non-zero variance	Must pass
Clean universe	118 of 145 stocks

Survivorship bias: Universe defined as of April 2026 — stocks that delisted or were acquired between 2015–2026 are under-represented. Results should be interpreted as *upper-bound* estimates of achievable historical performance.

Return Distribution — Key Statistics

Pooled monthly return distribution

(all 118 stocks $\times$ 135 months)

Statistic	Value
Mean	1.70%/month
Std. deviation	10.68%
Skewness	+0.68 (mild right skew)
Excess kurtosis	5.49 (fat tails)

Interpretation:

The positive skewness reflects occasional large positive returns (rallies). The fat tails ($\kappa \approx 5.5$ vs. Normal = 0) confirm that extreme co-movement events — exactly what we aim to study — occur far more often than a Gaussian model would predict.

The data matrix

$$\mathbf{X} \in \mathbb{R}^{N \times D}$$

- **Rows** = time periods ($N = 2781$ daily, $N = 135$ monthly)
- **Columns** = stocks ($D = 118$)
- **Entries** = excess returns $\tilde{r}_{i,t}$
- Before PCA: cross-sectionally standardised

The ratio $N/D \approx 23.6$ for the daily panel sounds comfortable, but a significant fraction of the spectrum of the sample covariance is still governed by estimation noise — the Marchenko–Pastur framework makes this precise.

Baseline Factor Discovery — Variance Explained

Monthly cross-sectional PCA

($N = 135$, $D = 118$)

PC	Indiv. EVR	Cumul. EVR
1	4.67%	4.67%
2	3.81%	8.48%
3	3.13%	11.61%
4	2.52%	14.13%
5	2.44%	16.57%
6	2.35%	18.92%
7	2.11%	21.03%
8	1.98%	23.01%
9	1.94%	24.95%
10	1.79%	26.74%

Key observations:

- PC1 explains only 4.67% — *after* removing the market component, there is no single dominant axis
- Top-5 PCs together explain 16.6%; remaining 83.4% is idiosyncratic
- Gradual eigenvalue decay — *no sharp elbow* — consistent with many weakly correlated style/sector factors
- Daily PCA gives virtually the same structure (PC1: 4.33%, top-5: 14.4%) confirming frequency-independence of cross-sectional structure

Compare: raw covariance PCA (without cross-sectional standardisation) would yield $PC1 \approx 20\text{--}30\%$. The low values here are *by design*.

Baseline Factor Discovery — Economic Interpretation

PC1 (4.67%) — Domestic Cyclical vs. Defensive Growth

52% of stocks carry *positive* loadings — a near even split.

High: ELECON, INOXWIND, CANBK (mid-cap industrials/energy)

Low: INFY, HCLTECH, DRREDDY, SUNPHARMA (IT/pharma)

PC2 (3.81%) — Technology vs. Commodities

High: COFORGE, PERSISTENT, KPITTECH (mid-cap IT)

Low: ADANIGREEN, HINDPETRO, IOC (energy/metals)

Consistent with divergent sensitivity to USD/INR and crude oil.

PC3 (3.13%) — Green Energy & Digital vs. Traditional Finance

High: ADANIGREEN, INOXWIND, COFORGE (renewables/digital)

Low: PGEL, BAJFINANCE, MUTHOOTFIN (NBFC/logistics)

PC4 (2.52%) — PSU Industrial & Pharma vs. Consumer Finance

High: BALRAMCHIN, LUPIN, BHEL (PSU/pharma)

Low: TANLA, BAJFINANCE, AUBANK (consumer finance)

These are **economic interpretations** of loading patterns — not mechanical outputs. They require corroboration via fundamental data (book-to-market, momentum).

COVID-19 Structural Break — Rolling PC1 Variance

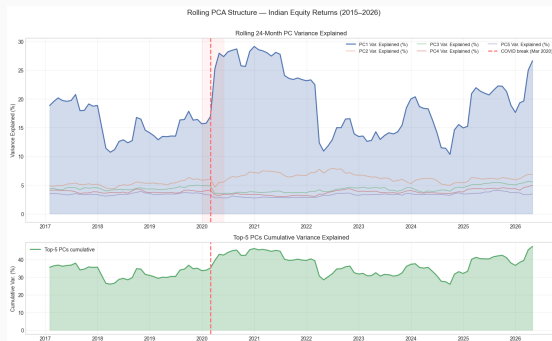
Setup: 24-month rolling PCA on 106 stocks present throughout 2015–2026, using within-window time-series standardisation.

Results:

- Pre-COVID mean PC1 variance: **15.9%**
- Post-COVID mean PC1 variance: **19.8%**
- Overall mean (112 windows): 18.5%
- **Change: +3.9 percentage points**

Interpretation:

The COVID shock forced stocks that previously moved on idiosyncratic fundamentals to move more in lockstep. This increased synchronisation has not reverted — suggesting a *permanent* shift in market microstructure, not a transient crisis response.



Chow Test Results — Statistical Evidence

Table: Chow Test for PC1 Variance Trend Break

Test quantity	Value
<i>F</i> -statistic	22.93
<i>p</i> -value	< 0.0001
Degrees of freedom	(2, 108)
Pre-break observations	38
Post-break observations	74
Decision ($\alpha = 0.05$)	Reject H_0

What this means: We reject the null that the slope and intercept of the PC1 variance time-trend are the same across the break.

The directional reversal is key

- **Pre-COVID trend:** slightly *declining* PC1 variance — the Indian market was maturing; sector differentiation was increasing
- **Post-COVID trend:** *rising* PC1 variance — synchronisation increased and continued to rise

This is not merely a level shift — it is a **directional reversal of the trend**, which constitutes the strongest possible parametric evidence of a regime change.

Caveat: The break date (March 2020) is specified *a priori*, not identified endogenously. This is a limitation; an endogenous break test (e.g. Andrews–Quandt) is left for future work.

Procrustes Analysis Results — Loading Composition

Pre vs. Post-COVID comparison

Separate full-sample PCAs on:

- Pre-COVID sub-sample: $N = 61$ months (Jan 2015 – Feb 2020)
- Post-COVID sub-sample: $N = 74$ months (Apr 2020 – Apr 2026)

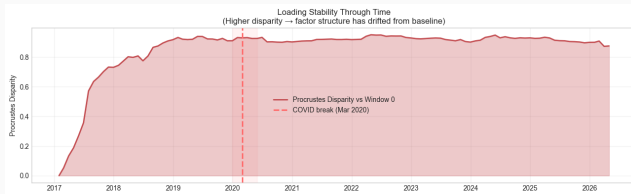
After Procrustes alignment of the $K = 5$ leading eigenvector matrices:

$$\delta_{\text{pre} \rightarrow \text{post}} = 0.782$$

Loading stability time series

Procrustes disparity of each rolling window vs. baseline (ending Feb 2017):

- Gradual drift 2015–2019: slow secular change
- **Sharp jump:** late 2019 → peaking Mar–Apr 2020
- **No reversion:** disparity stays elevated



Summary of Evidence

Table: Three Convergent Lines of Evidence for a Structural Break at COVID-19

Test	Result	Interpretation
Chow test (linear trend)	$F = 22.93, p < 0.001$	PC1 variance trend <i>reversed direction</i> at March 2020
Procrustes disparity (pre → post)	$\delta = 0.782$	Loading composition changed very substantially; the stocks driving factors are fundamentally different
Loading stability series	No reversion to pre-2020 level	Break is <i>permanent</i> within sample, not a transient perturbation
Pre vs. post mean PC1 variance	15.9% → 19.8% (+3.9 pp)	Permanently elevated market-wide co-movement

All three methods agree: COVID-19 produced a permanent structural break in the Indian equity factor structure — both in the *strength* and in the *composition* of systematic co-movement.

Conclusions — What Did We Find?

1. **Low idiosyncratic explained variance.** After removing the market component via cross-sectional standardisation, the leading PCs explain modest variance fractions (PC1: 4.7%, top-5: 16.6%). This is a *feature* — it confirms that the Indian cross-section has a rich multi-factor idiosyncratic structure beyond the market return.
2. **Four interpretable style and sector factors.** PC1–PC4 correspond economically to:
 - (i) Domestic cyclical vs. defensive growth
 - (ii) Technology vs. commodities
 - (iii) Green energy & digital vs. traditional finance
 - (iv) PSU industrials & pharma vs. consumer finance
3. **Permanent structural break at COVID-19.** Rolling PCA, Chow test ($F = 22.93$, $p < 0.001$), and Procrustes alignment ($\delta = 0.782$) all independently confirm that the factor structure changed materially and permanently around March 2020.

Recommendations

For Policymakers (SEBI / FSB)

- **Systemic risk monitoring:** Rolling PC1 variance is a simple, model-free co-movement indicator. Include it in quarterly financial stability reports — a rising PC1 share signals increasing tail dependence.
- **Stress-test correlation assumptions:** Risk models calibrated on pre-2020 matrices will underestimate post-COVID tail dependence. Regulators should require institutions to stress-test with post-2020 loading matrices.
- **Promote sector diversification:** Support sectors with persistently low PC1 loadings (domestic manufacturing, healthcare, digital infrastructure) that provide genuine portfolio diversification.

For Investors & Fund Managers

- **Dynamic factor exposure:** Factor loadings are not stable ($\delta = 0.782$). Re-estimate risk models *at least annually*; pre-2020 factor models provide misleading exposures today.
- **PC1 variance as regime indicator:** PC1 > 20% signals a high-correlation regime — a natural trigger to reduce gross exposure or increase hedging.
- **Robust covariance estimation:** Ledoit–Wolf shrinkage materially stabilises PCA-based factor estimates in 24-month rolling windows. Practitioners relying on raw sample covariance matrices are likely overfitting to noise.

Limitations & Future Work

Limitations of this study

- **Survivorship bias:** Universe defined as of April 2026 — delisted/merged firms are excluded, biasing performance estimates upward.
- **Window overlap:** 24-month rolling windows cannot cleanly isolate the acute COVID period from the slow-building NBFC crisis of 2018–19.
- **Informal factor labelling:** Economic interpretations of loadings are based on stock names, not formal characteristics matching (book-to-market, momentum).
- **Pre-specified break date:** Chow test is conditional on March 2020; endogenous break identification is not performed.

Directions for future work

- **Endogenous break tests:** Andrews–Quandt supremum- F test to identify the break date without assuming March 2020.
- **Robust replication:** Replicate on a universe of all NSE-listed stocks (including delisted) to remove survivorship bias.
- **Characteristics matching:** Formally attribute PCA factors to Fama–French characteristics (size, value, profitability, investment) using cross-sectional regressions.
- **Dynamic factor models:** Replace rolling PCA with a Kalman filter–based state-space factor model for smoother, probabilistic tracking of time-varying loadings.

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